

EXPERIMENT - 2 REPORT

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EE19B108 , Batch 1

OBJECTIVES:

To implement basic arithmetic and logical manipulation programs using Atmel Atmega8 microcontroller in assembly program emulation, including addition, multiplication and comparison.

HARDWARE AND SOFTWARE USED:

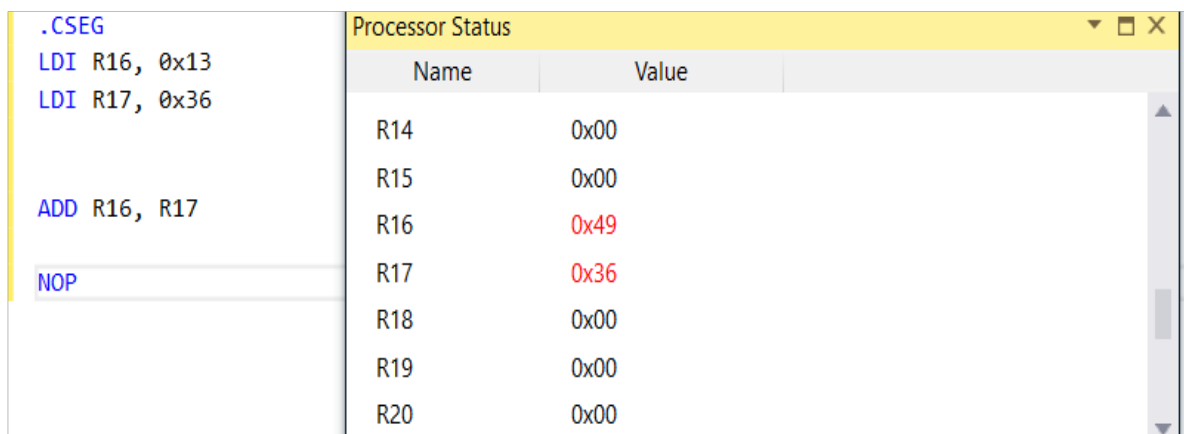
A Windows PC with Microchip Studio IDE (Integrated Development Environment) for developing and debugging AVR & SAM microcontroller applications.

Code for the first problem (8-bit addition):

```
CSEG          ; Code Segment

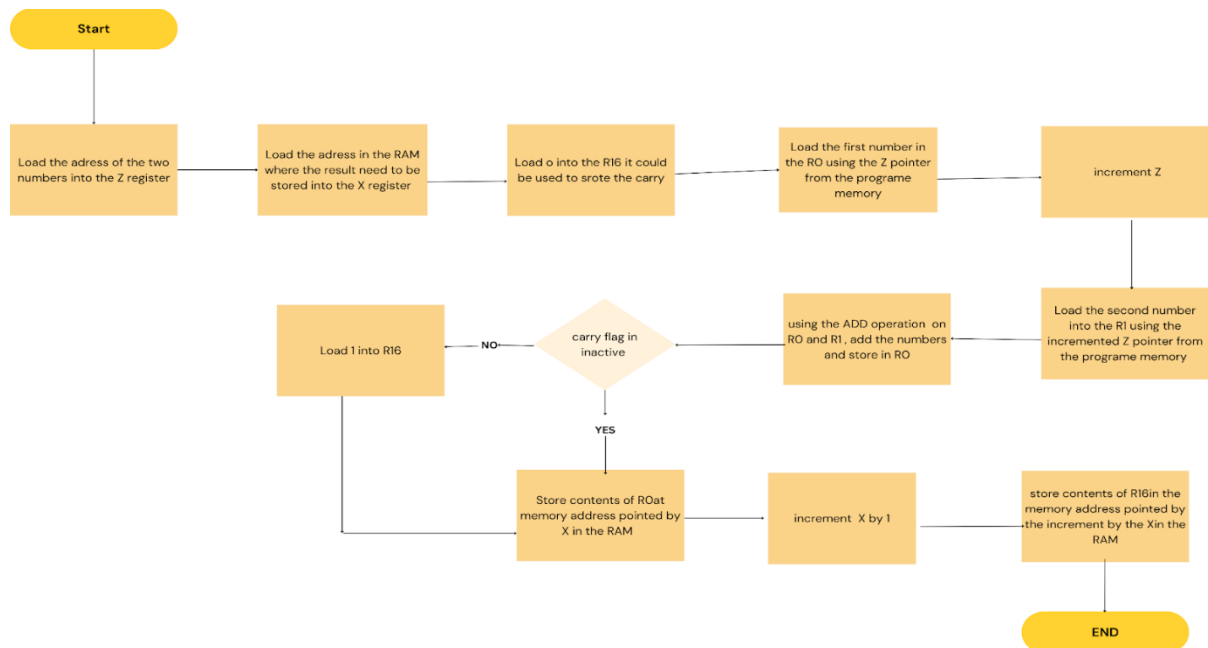
LDI R16, 0x13  ; Load first number into R16
LDI R17, 0x36  ; Load second number into R17
ADD R16, R17   ; Add both numbers , store in R16

NOP
```



Processor Status	
Name	Value
R14	0x00
R15	0x00
R16	0x49
R17	0x36
R18	0x00
R19	0x00
R20	0x00

Final output result stored in R16 register(as shown in above image)



Code for the second problem(16-bit addition using 8-bit processor):

.CSEG ; Code Segment

```

LDI R16, 0x12 ; Load lower byte of first number into R16
LDI R17, 0x34 ; Load higher byte of first number into R17
LDI R18, 0x56 ; Load lower byte of second number into R18
LDI R19, 0x78 ; Load higher byte of second number into R19
ADD R16, R18 ; Add lower bytes, store in R16
ADC R17, R19 ; Add higher bytes with carry, store in R17

```

NOP

```

; Replace with your app
start:
.CSEG
LDI R16, 0x12
LDI R17, 0x34
LDI R18, 0x56
LDI R19, 0x78

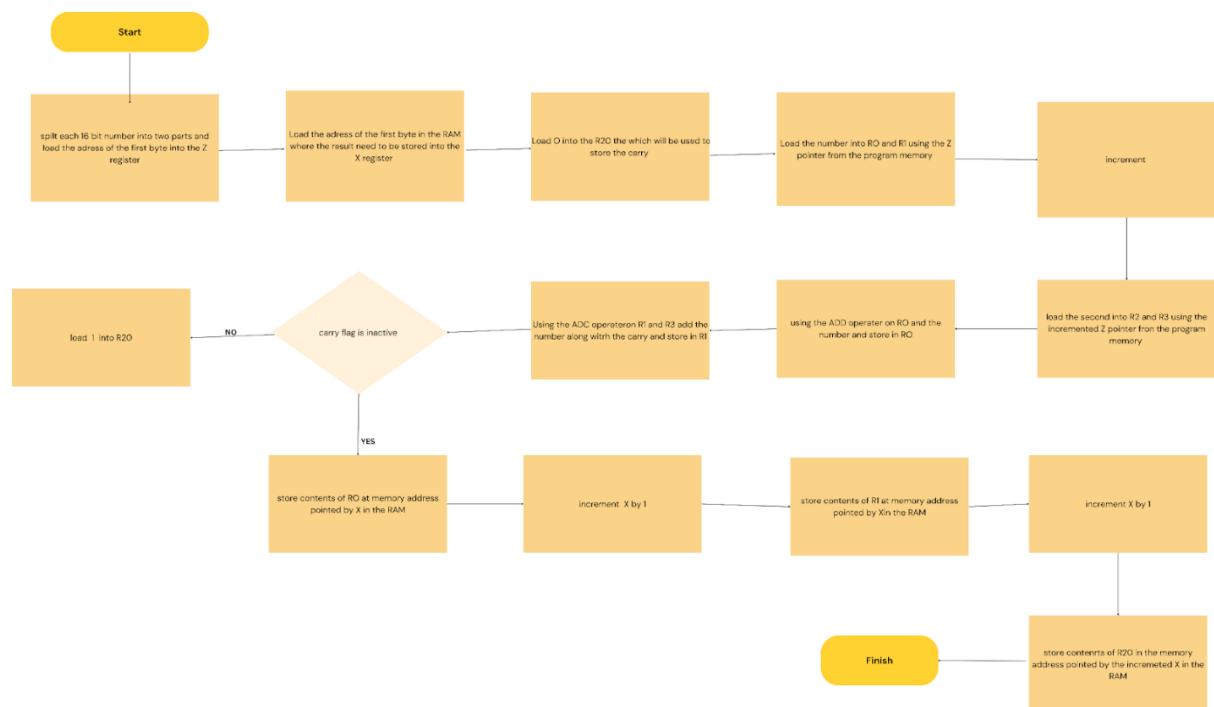
ADD R16, R18
ADC R17, R19

NOP

```

Processor Status	
Name	Value
R14	0x00
R15	0x00
R16	0x68
R17	0xAC
R18	0x56
R19	0x78
R20	0x00

bytes and higher bytes of output is stored in R16 and R17 respectively (as shown in above image)



Code for the third problem(8-bit multiplication):

.CSEG ; Code Segment

LDI R16, 0xA ; Load first number into R16
 LDI R17, 0xB ; Load second number into R17
 MUL R16, R17 ; Multiply R16 and R17, result in R1:R0

NOP

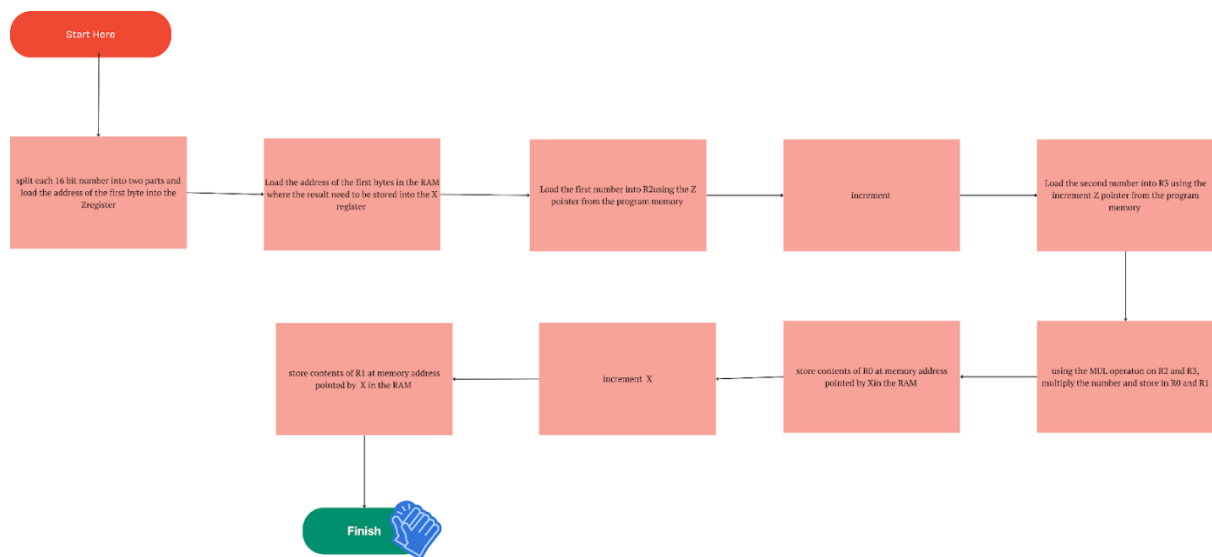
```

; Replace with your assembly code
start:
.CSEG
LDI R16, 0xA
LDI R17, 0xB
MUL R16, R17

NOP
        
```

Processor Status	
Name	Value
Registers	
R00	0x6E
R01	0x00
R02	0x00
R03	0x00
R04	0x00
R05	0x00

lower bytes of output is stored in R0 and higher bytes of output is stored in R1



Code for the fourth problem (largest of the numbers given):

.CSEG ; Code segment starts

LDI R16, 0x25 ; Load immediate value 0x25 into register R16
LDI R17, 0x37 ; Load immediate value 0x37 into register R17
LDI R18, 0x54 ; Load immediate value 0x54 into register R18
LDI R19, 0x12 ; Load immediate value 0x12 into register R19
LDI R20, 0x62 ; Load immediate value 0x62 into register R20

CP R16, R17 ; Compare R16 with R17
BRGE Check_R18 ; Branch to Check_R18 if R16 >= R17
MOV R16, R17 ; If R16 < R17, move value of R17 into R16

Check_R18:
CP R16, R18 ; Compare R16 with R18
BRGE Check_R19 ; Branch to Check_R19 if R16 >= R18
MOV R16, R18 ; If R16 < R18, move value of R18 into R16

Check_R19:
CP R16, R19 ; Compare R16 with R19
BRGE Check_R20 ; Branch to Check_R20 if R16 >= R19
MOV R16, R19 ; If R16 < R19, move value of R19 into R16

```

Check_R20:
CP R16, R20      ; Compare R16 with R20
BRGE Set_All    ; Branch to Set_All if R16 >= R20
MOV R16, R20     ; If R16 < R20, move value of R20 into R16

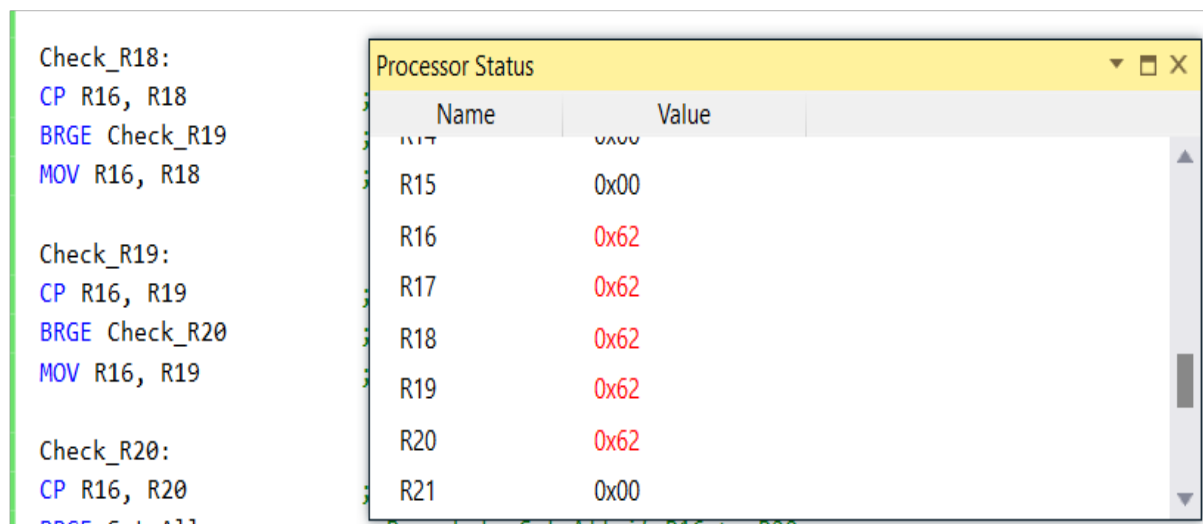
```

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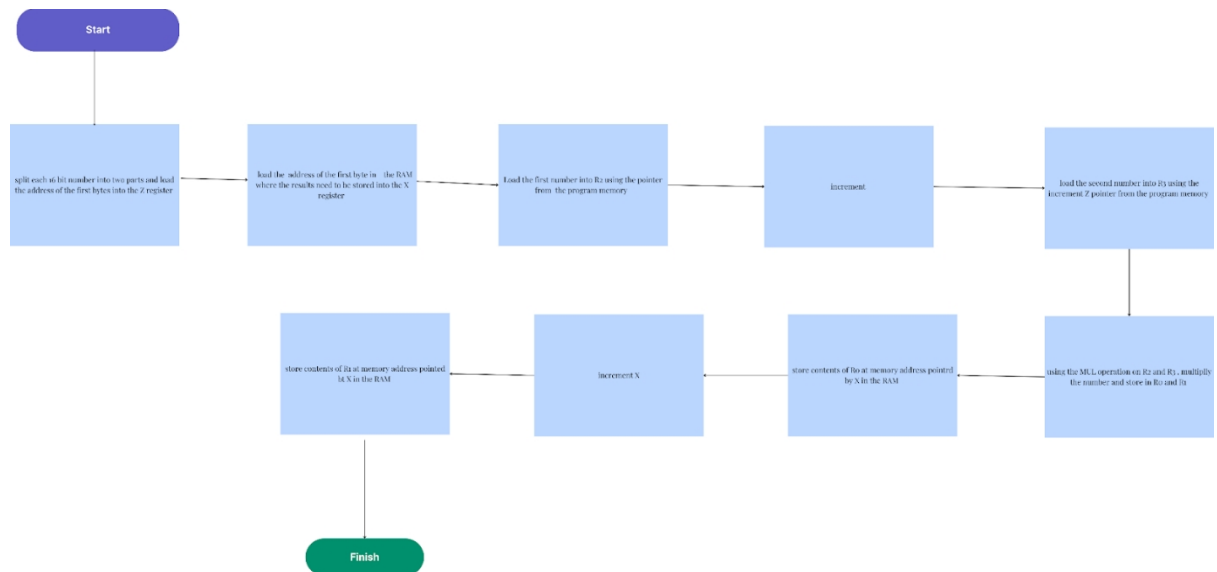
Set_All:
MOV R17, R16     ; Set R17 = R16 (store the smallest value in R17)
MOV R18, R16     ; Set R18 = R16 (store the smallest value in R18)
MOV R19, R16     ; Set R19 = R16 (store the smallest value in R19)
MOV R20, R16     ; Set R20 = R16 (store the smallest value in R20)

```

NOP



Largest number among the other numbers is stored in all registers R16,R17,R18,R19 and R20 (as shown in above image)



RESULTS AND INFERENCE :

Addition, subtraction, multiplication, and comparison of 8-bit / 16-bit binary numbers are demonstrated through emulation of Atmega8 assembly programming.